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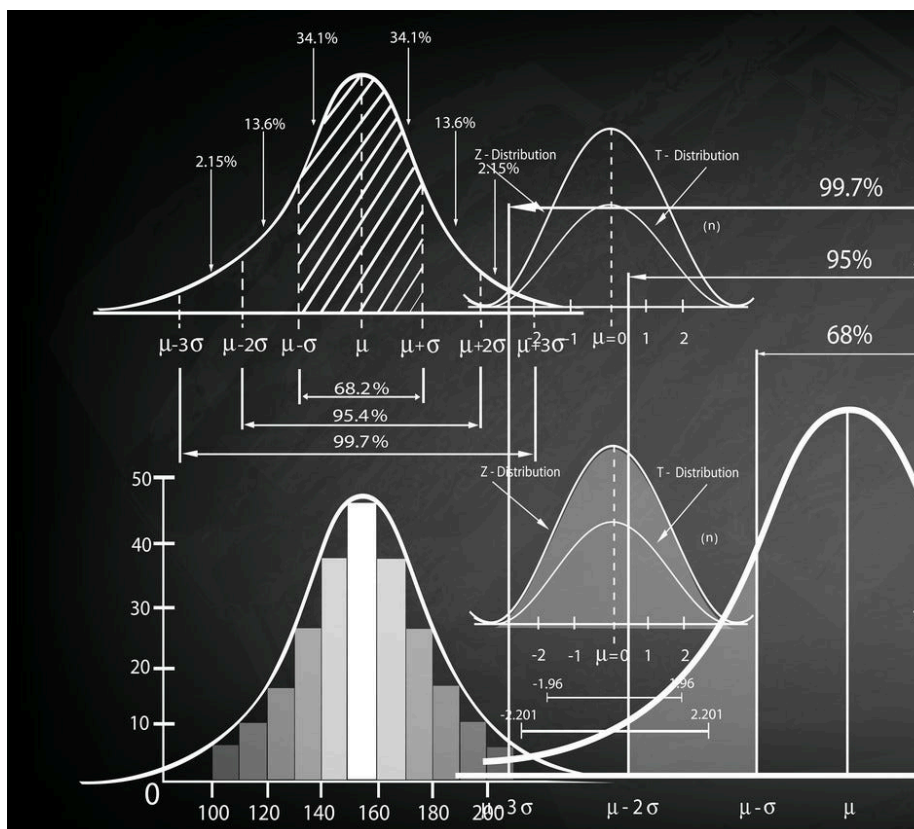
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Probability Learning II: How Bayes Theorem is applied in Machine Learning

Learn how Bayes Theorem is in Machine Learning for regression!

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In the previous post we saw **what Bayes' Theorem is**, and went through an easy, intuitive example of how it works. You can find this post [here](#). If you don't know what Bayes' Theorem is, and you have not had the pleasure to read it yet, I recommend you do, as it will make understanding this present article a lot easier.

In this post, we will see the **uses of this theorem in Machine Learning**.

Before we start, here you have some additional skyrocket your Machine Learning career:

Awesome Machine Learning Resources:

- For learning resources go to How to Learn Machine Learning
- For professional resources (jobs, events, skills) go to Machine Learning Career

Ready? Lets go then!

Bayes' Theorem in Machine Learning

As mentioned in the previous post, Bayes' theorem is used to **gradually update our knowledge** on *something* based on new evidence or that about that *something*.

Generally, in **Supervised Machine Learning**, when we train a model the **main building blocks** are a set of data points that contain **features** (the attributes that define each data point), **the labels** of such data point (the numerical or categorical tag which we later want to predict on new data), a **hypothesis function** or model that links such features to the corresponding labels. We also have a **loss function** that measures the difference between the predictions of the model and the actual labels.

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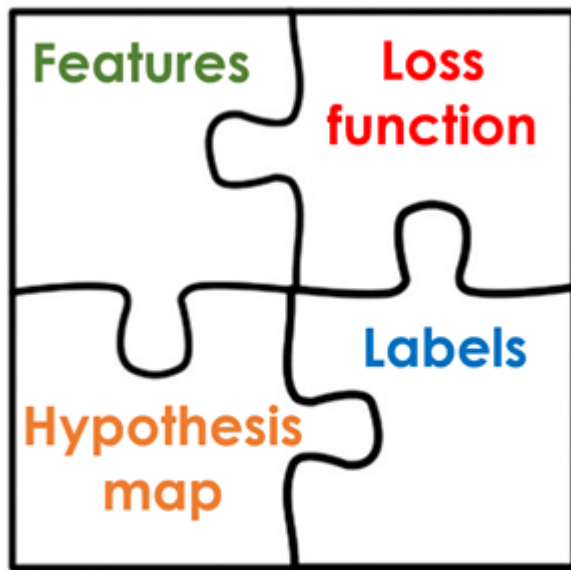
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labels which we want to reduce to achieve the best possible results.



Main elements of a supervised Learning Problem

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These supervised Machine Learning problems can be divided into **two main categories: regression**, where we want a number or **numeric value** associated with some example (the price of a house), and **classification**, to assign the data point to a **certain category** (if an image shows a dog or a cat).

Bayes' theorem can be used in both regression and classification.

Lets see how!

Bayes' Theorem in Regression

Imagine we have a very **simple set of data**, which is the **temperature of each day** of the year in a certain city (the **feature** of the data points), and the **number of days** in that month (the **label**).

sold by a local shop in that area every single day (the **label** of the data points).

By making a **very simple model**, we could **see if these two are related**, and if they are, then use this model to **make predictions** in order to stock up on water bottles depending on the temperature and never run out of stock, or avoid having too much inventory.

We could try a very simple **linear regression model** to see if these variables are related. In the following formula, y describes this linear model, y is the target label (the number of water bottles in our example), **each of the θ s is the model** (the slope and the cut with the y-axis), and x is our feature (the temperature in our example).

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$$y = \theta_0 +$$

Equation describing a linear model

The goal of this training would be to **reduce the cost function**, so that the predictions that the model makes for known data points, are close to the actual values of such data points.

After having trained the model with the available data, we would get a value for both of the **θ s**. This training can be done using an **iterative process** like gradient descent or a **probabilistic method** like Maximum Likelihood. We would just **have ONE single value** for each one

In this manner, when we get **new data without a label** (new temperature forecasts) as we know the value of the **θ s**, we could just use this simple equation to obtain the wanted **Y s** (number of water bottles needed for each day).

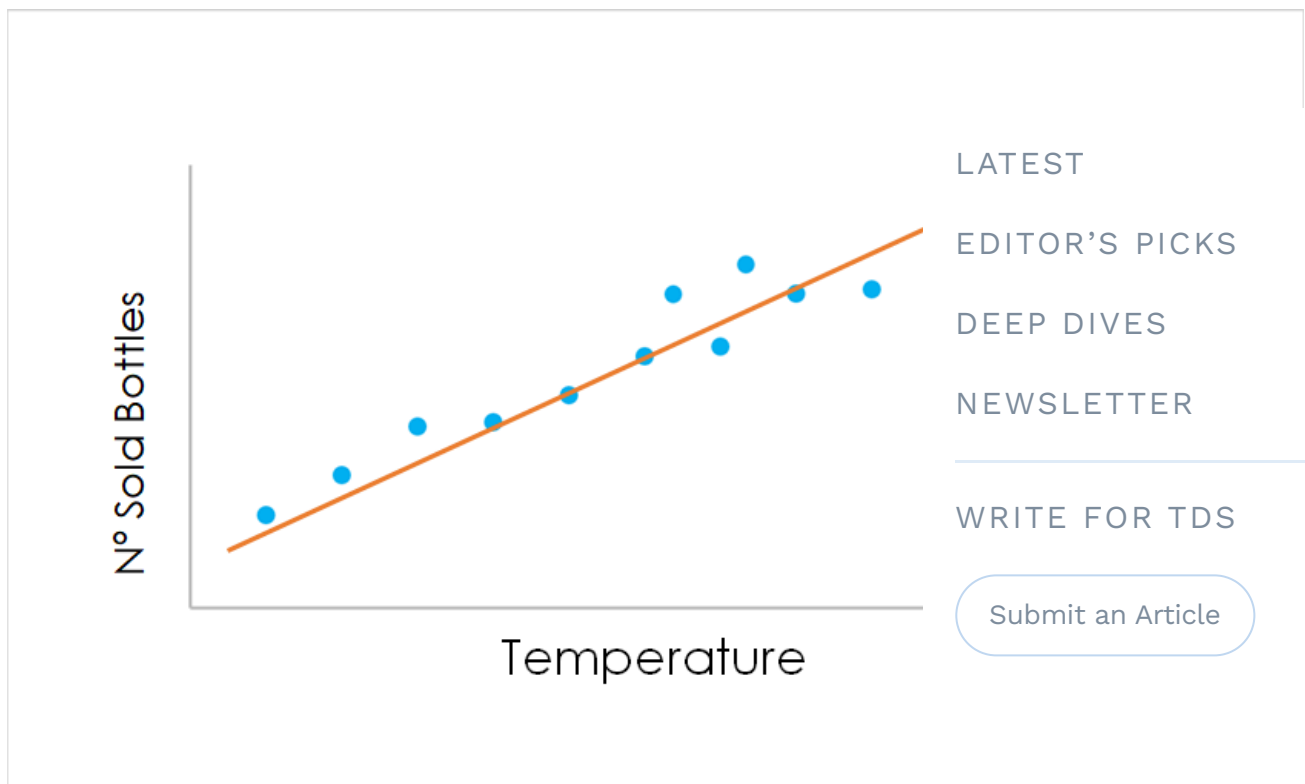


Figure of an uni-variate linear regression. Using the initial blue data points, we fit a line to these points, and then when we get a new temperature we can easily calculate the number of bottles sold that day.

When we use Bayes' theorem for regression, in **of the parameters** (the θ s) of the model as having a unique value, we represent them **as** parameter **distribution**: the prior distribution of the parameters. The following figures show the generic Bayes formula and how it can be applied to a machine learning model.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Bayes formula

$$P(model|data) = \frac{P(data|model)P(model)}{P(data)}$$

Bayes formula applied to a machine learning model

The idea behind this is that **we have some previous knowledge of the parameters of the model** before we have any actual data:

$P(model)$ is this prior probability. Then, **when we have data, we update the distribution of the parameters** making it the posterior probability **$P(model|data)$**

What this means is that **our parameter set** (the set of parameters) is not constant, but instead **has its own distribution**. This represents our previous knowledge (from experts for example, or from other similar works) **we make a first hypothesis** about the distribution of the parameters of our model. Then as we train our model on **data, this distribution gets updated** and grows more precise (as we practice the variance gets smaller).

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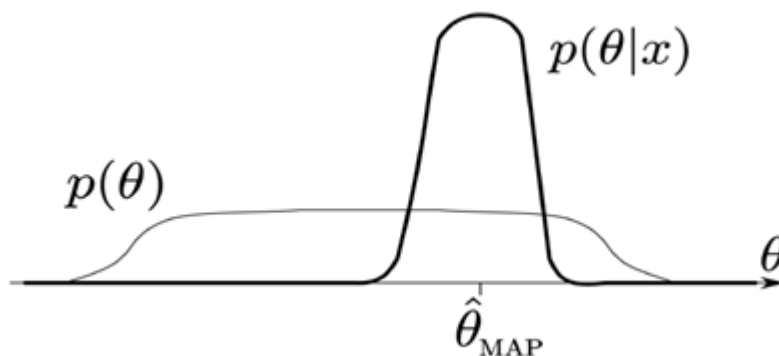


Figure of the a priori and posteriori parameter distributions. θ_{MAP} is the maximum a posteriori estimate, the value of θ that we would then use in our models.

This figure shows the **initial distribution of the model $p(\theta)$** , and how as we add more data this **updates**, making it grow more exact to **$p(\theta|x)$** , the distribution of the parameters given this new data. The θ here is equivalent to the θ in the equation above.

formula shown above, and the $\mathbf{x}$ here is equivalent to the *data* in such formula.

Bayes' formula, as always, tells us **how to go from the prior to the posterior probabilities**. We do this in an iterative process as we get more and more data, having the **posterior probabilities become the prior probabilities for the next iteration**. Once we have trained the model with enough data, to choose the final parameters we would search for the **Maximum a Posteriori (MAP) estimation to use a concrete set of values for the parameters of the model**.

This kind of analysis **gets its strength from the distribution**: if we do not have any previous information, we can't make any assumption about it, other probabilistic approaches like Maximum Likelihood are better.

However, **if we have some prior information about the distribution of the parameters the Bayes' approach is very powerful**, specially in the case of having **unlabeled data**. In this case, as we are not building the model by calculating its parameters from scratch using training data, but rather using some kind of previous knowledge (the prior distribution for these parameters, **this previous knowledge makes the parameters more robust and less affected by inaccurate data**).

I don't want to get very technical in this part, but behind all this reasoning is beautiful; if you want to learn more, don't hesitate and email me to jaimezorno@gmail.com or [find me on LinkedIn](#).

Bayes' Theorem in Classification

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We have seen how Bayes' theorem can be used for regression, by estimating the parameters of a linear model. The same reasoning could be applied to other kind of regression algorithms.

Now we will see how to use Bayes' theorem for classification. This is known as **Bayes' optimal classifier**. The reasoning now is very similar to the previous one.

Imagine we have a classification problem with n classes. The thing we are after here is **the class probabilities** $p(w_i)$. Like in the previous regression case, we also have a relationship between prior and posterior probabilities, but in this case we are interested in **class probabilities** $p(w_i)$ and **posterior class probabilities** $p(w_i|x)$ using data or observations $p(w_i|x)$.

$$P(w_i|x) = \frac{P(x|w_i)P(w_i)}{P(x)}$$

Bayes formula used for Bayes' optimal classifier

Here $P(x)$ is the **density function** common to all classes. $P(x|w_i)$ is the **density function of the data point** for class w_i , and $P(w_i)$ is the prior distribution of class w_i , calculated from the training data, assuming a normal distribution and calculating a **mean vector for each class** and **variance of the features** of the data points belonging to that class. Prior class distributions $P(w_i)$ are estimated based on **knowledge**, expert advice or previous works, like in the previous example.

Lets see an example of how this works: Imagine we have the height of 34 individuals: **25 males (blue)** and 9 females (orange) and we get a **new** height **observation** of 172 cm.

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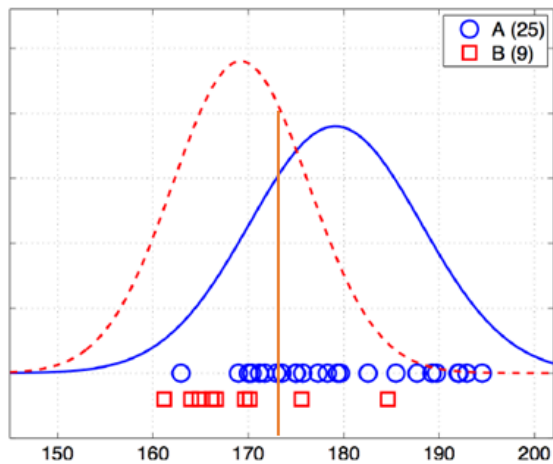
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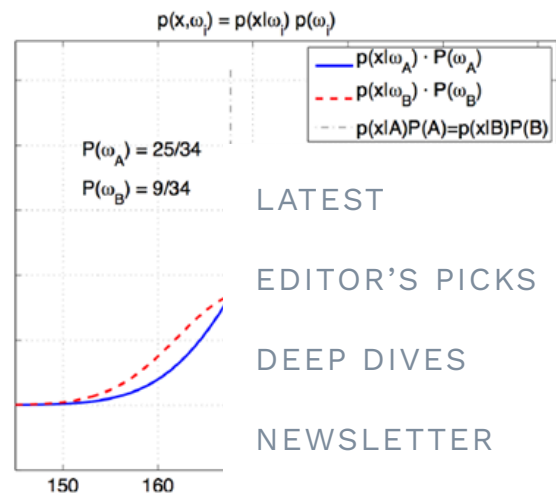
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classify as male or female. The following figure represents the predictions obtained using a **Maximum likelihood classifier** and a **Bayes optimal classifier**.



Maximum Likelihood classifier

Predicted class: **Female**



Bayes optimal classifier

On the left, the training data for both classes with their estimated normal distributions. On the right, the Bayes optimal classifier, with prior class probabilities $p(\omega_A)$ of male being 25/34 and $p(\omega_B)$ of female being 9/34.

In this case we have used the **number of samples** as the **prior knowledge** for our class distributions. In the previous example we were doing this same differentiation based on height and gender for a specific country, and we knew that people from that country are specially tall, and also knew the mean height. In this case, we could have used this **information to build our prior distributions**.

As we can see from the example, using these **prior distributions** leads to different results than not using them. If our previous knowledge is of high quality (or otherwise, if we can use it), these predictions should be **more accurate** than trials that don't incorporate this information.

After this, as always, as we get **more data** these predictions **should get updated** to reflect the knowledge of the new data.

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data.

As in the previous case, I don't want to get too technical, or extend the article too much, so I won't go into the mathematical details, but **feel free to contact me if you are curious about them.**

Conclusion

We have seen **how Bayes' theorem is used in NLP** both in **regression** and **classification**, to incorporate knowledge into our models and improve them.

In the following post we will see how **simplified Bayes' theorem** are one of the most used techniques in **Language Processing** and how they are applied in world use cases like spam filters or sentiment analysis. check it out [follow me on Medium](#), and stay tuned.

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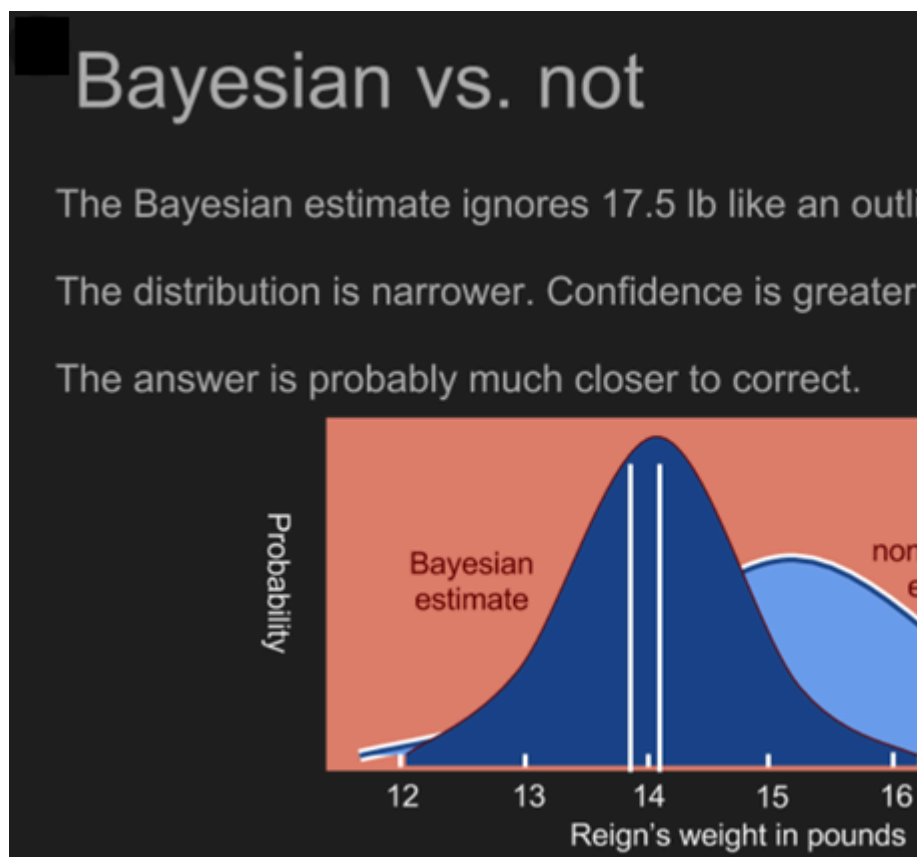
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Another example of Bayesian classification

That is all, I hope you liked the post. Feel Free to connect with me on [LinkedIn](#) or follow me on Twitter at [@jaimezorno](#). Also, you can take a look at my other posts on Data Science and Machine Learning [here](#). Have a good read!

Additional Resources

In case you want to go more in depth into Bayesian Learning, check out these other resources:

- [How Bayesian Inference works](#)
- [Bayesian statistics Youtube Series](#)
- [Machine Learning Bayesian Learning slides](#)
- [Bayesian Inference](#)
- [Statistics and Probability online courses](#)

For further resources on Machine Learning and check out the following repository: [How to Learn Machine Learning](#)! For career resources (jobs, events, skills) check out [Algen.co](#) – A career community for Data Scientists and Machine Learning Engineers and as always, contact me if you have any questions. Have a fantastic day and keep learning!

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